

Multimedia Databases

Media - Text, Video, Audio

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- 1 Characters and their classification
 - 1.1 Media type „Text“
 - 1.2 XML
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 - 2.2 Shot segmentation
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 - 2.5 Interactive videos
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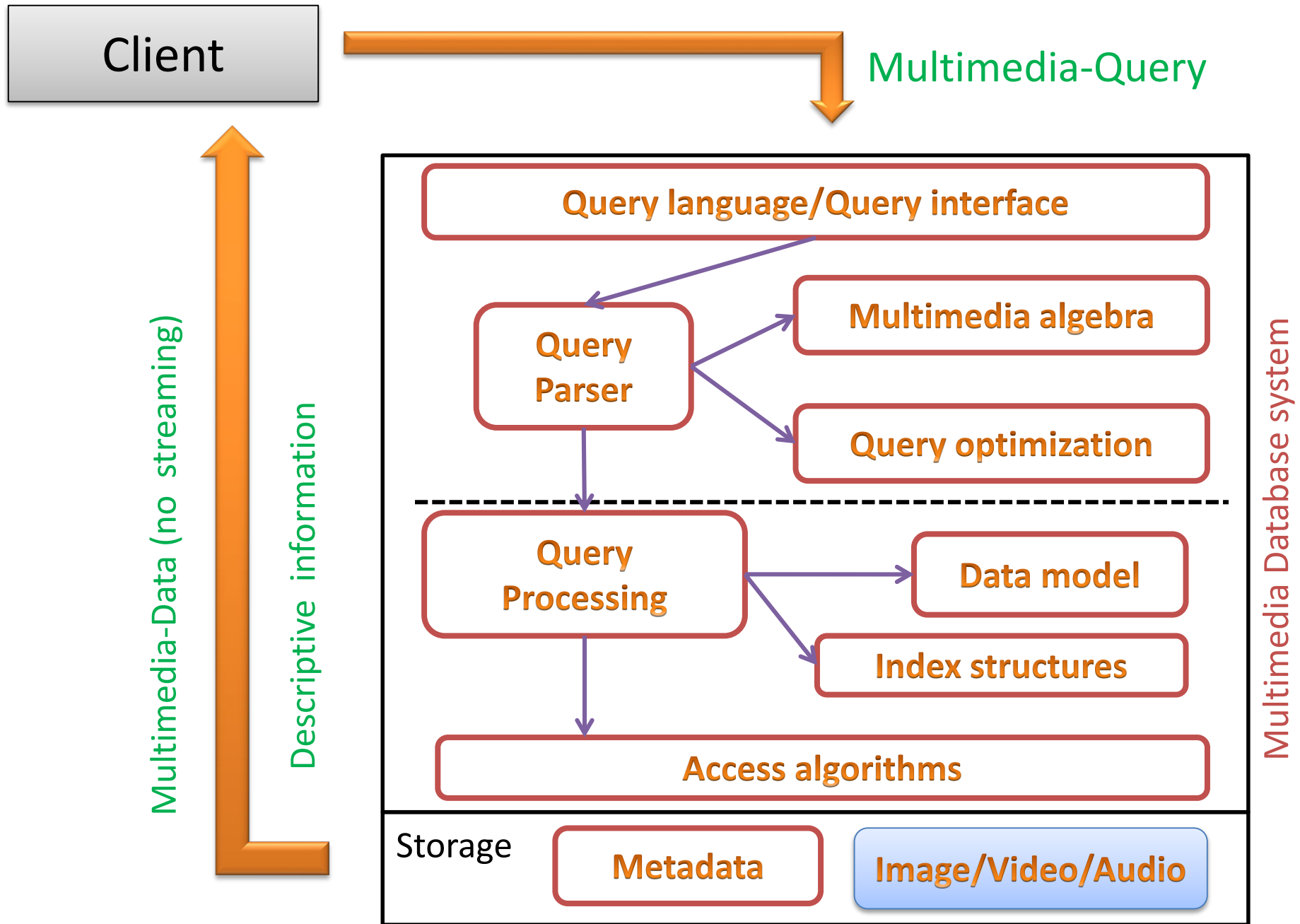
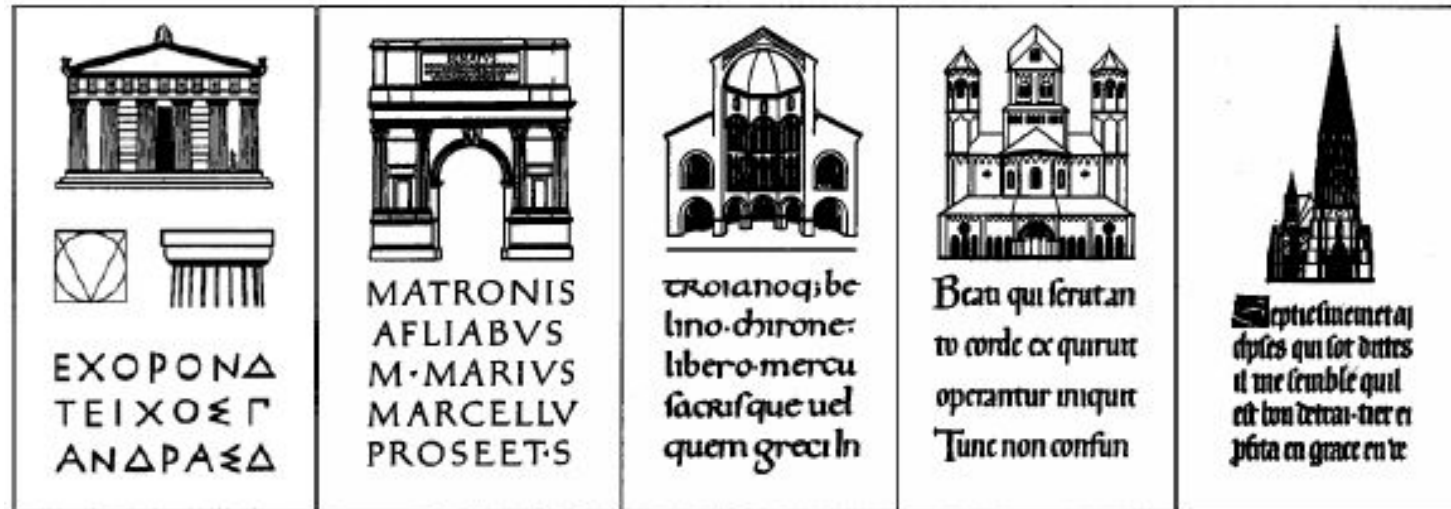


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History of Characters



Greek
500 b.c.
Geometri
c forms

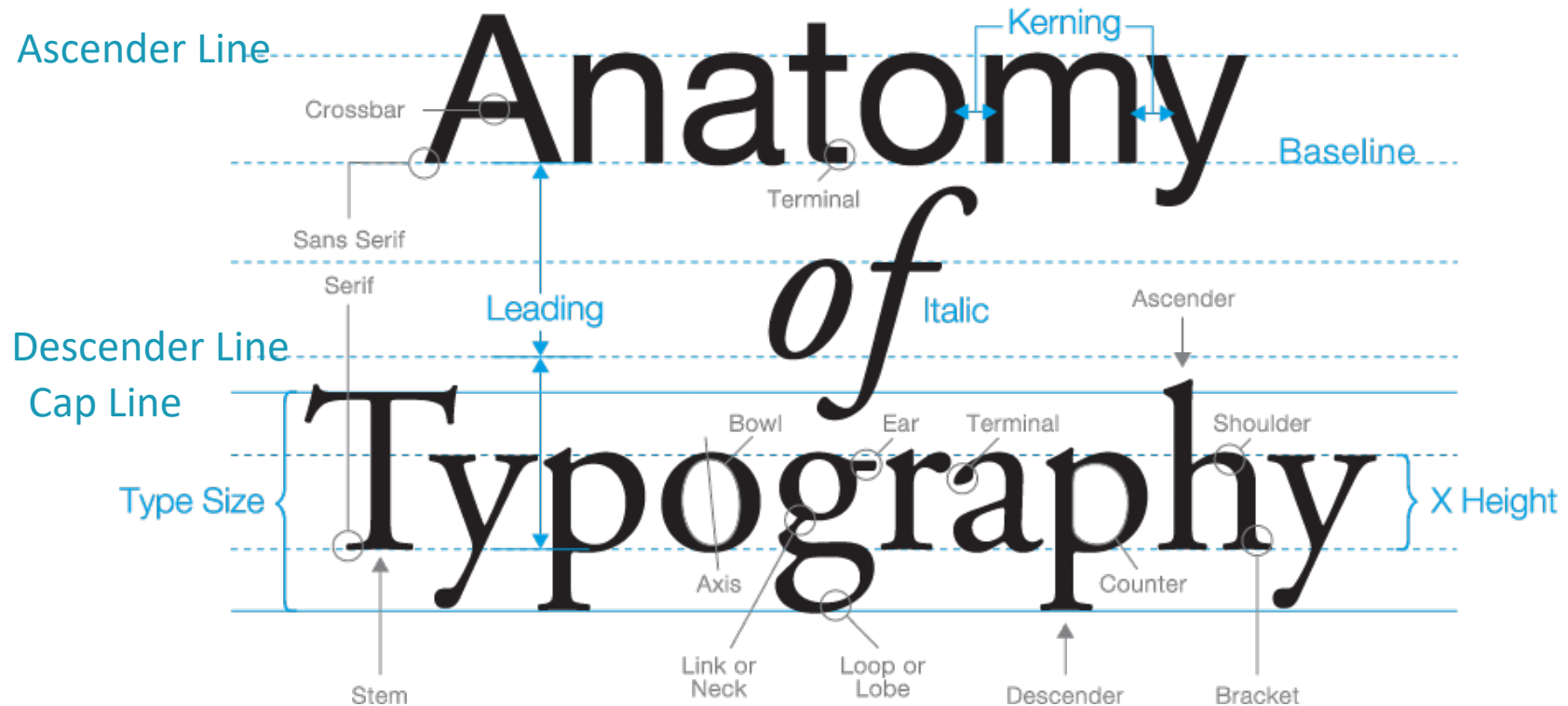
Roman
100
Capitalis
Quadrata

Carolingian
800
Small letter
fonts

Romanic
1100

Gothic
1250
Textura

Typographic Measurements



- Font Size usually measured in Big Points (pt, bp), where $1\text{pt} = 1/72 \text{ inch} = 0.3527 \text{ mm}$
- Relative Measures
 - 1 ex = height of glyph
 - 1 em = width of glyph

Distinguishing Features of Characters

- Glyph width:
 - Monospaced (e.g. I M) vs. proportional (e.g. I M)

- Serif/Sans-serif

Serifenbehaftet
Serifenlos

- Combination of glyphs

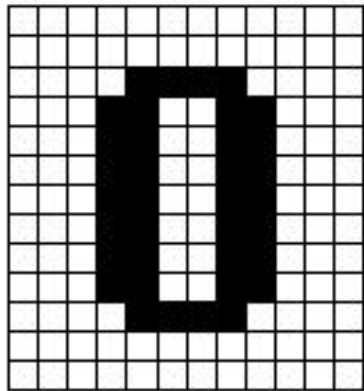
- Kerning
- Ligatures

AE → *Æ* *ij* → *ij*
ae → *æ* *st* → *ſt*
OE → *Œ* *ft* → *ft*
oe → *œ* *et* → *&*
ff → *ff* *fs* → *ß*
fi → *fi* *ffi* → *ffi*

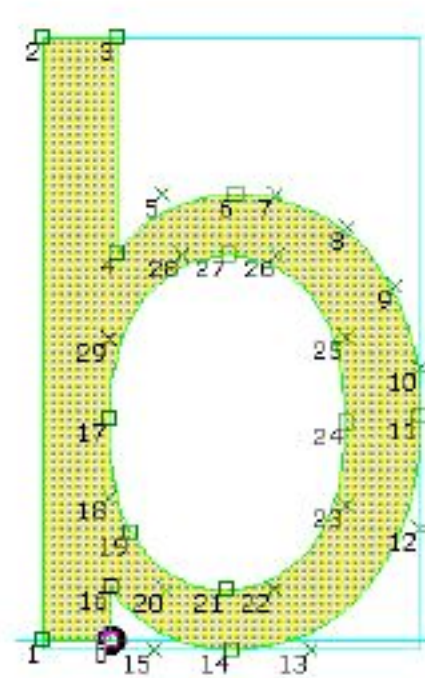
Typo Typo

Character Representation

Bitmapped fonts



Outlined fonts



© Slides partially borrowed from Prof. Dr.-Ing. Eckehard Steinbach, TU München

Bitmap Representation

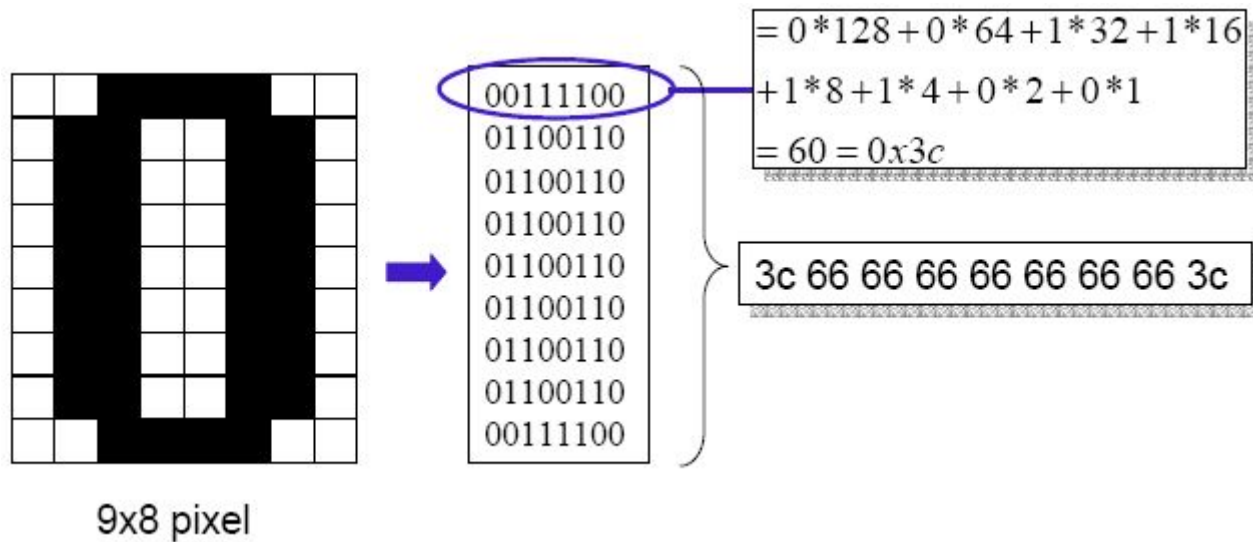


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Media type Text

- **REPRESENTATION**
 - ASCII
 - ISO character sets
 - Markup text
 - Structured text
 - Hypertext
- **OPERATIONS**
 - Character and string operations
 - Editing
 - Formating
 - Pattern recognition and search
 - Sorting
 - Compression
 - Encryption
 - Language specific operations

Representation of Text

- ASCII (American Standard Code for Information Interchange)
 - 7-Bit-Code, American Standard Code for Information Interchange, 1 Bit free for special characters
 - ISO character sets
 - Extension of ASCII to accentuated characters, e.g. Latin-1
- Markup text
 - Separation of form and content
 - Special tags to specify the structure of presentation
- Structured text
 - Processing oriented representation based on suitable data structures,
 - e.g. trees
- Hypertext
 - Graph oriented combination of information units
 - Nodes, edges, links

e.g. XML

Operations for Text

- Character operations
 - e.g. comparison 'a' < 'b'
- String operations
 - e.g. comparison 'abc' < 'abd'
 - Concatenation
'abc' + 'def' → 'abcdef'
- Editing
 - Modification of content and form
 - Copy, insert, remove, etc.
- Formatting
 - Assignment of layout information
 - Editor & Formatter → WYSIWYG
 - Page description languages→
Bitmap-oriented display (Postscript)
- Pattern recognition and search
 - e.g. search and replace
 - Regular expressions
- Sorting
- Compression
 - e.g. Huffman-Code (cf. JPEG),
Reduction of 1/2 -1/3
- Encryption
 - Important for data transmission
 - DES (Data Encryption Standard),
Public-Key cryptography

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Extensible Markup Language (XML)

- Developed by the **World Wide Web Consortium** (www.w3.org) since 1996
- Extensible **Markup Language**
- Goal: ensure two main properties:
 - **Flexibility** and performance of the Standard Generalized Markup Language (**SGML**),
 - Wide **acceptance** of **HTML**

⇒ XML is defined as a **subset of SGML**.
- XML = markup language for the representation of **structured data**.
- XML is **platform-independent**.
- XML is an **open standard**, which enable the encoded documents to **self-describe their structure**.
- XML is thus an „**ideal medium**“ for the exchange of documents or **data** between users or software.

Well-formed XML

- Contrarily to HTML, XML document must fulfill a **hard criterion** in order to be validated (e.g. by a parser).
- XML documents that are not „well-formed“ must be **rejected** by users or software.
- An XML document is **well-formed** if it respects the **following conditions**:
 - An XML document may only have a **single root-element**.
 - All subsequent elements must be connected to the root-element in a single **hierarchical tree structure**.
 - An XML document must not contain **invalid characters**. This condition refers in particular to the character set specified in the prolog.

Validation of XML-documents

- Well-formedness is an important criterion when verifying XML documents
 - It guarantees that the document is conform to the **general syntactic rules** of the Extensible Markup Language.
- More advanced criterion: **validity**
 - Means that the structure of the XML document is evaluated for conformity to the **corresponding Document Type Definition**.
 - If **no DTD was specified** for an XML document, it **cannot be validated**.
 - Only well-formed XML documents may be validated.
 - (an XML-Schema may also be used instead of a DTD)

Basic-Technologies

eXtensible Markup Language (XML)

```
<?xml version="1.0" encoding="ISO-8859-1" ?>
- <song>
  <title>Aber bitte mit Sahne</title>
  <author type="music">Udo Jürgens</author>
  <author type="text">Eckart Hachfeld</author>
  <date>1976</date>
  <duration format="min:sec">3:07</duration>
- <text>
  - <verse>
    <line>Sie treffen sich täglich um viertel nach drei, ohoho, (oh yeah!)</line>
    <line>am Stammtisch im Eck, in der Konditorei, ohoho, (oh yeah!)</line>
    <line>und blasen zum Sturm auf das Kuchenbuffet, auf Schwarzwälder Kirch und
      auf Sahne-Baisers,</line>
    <line>auf Früchteeis, Ananas, Kirsch und Banane, ABER BITTE MIT SAHNE!</line>
  </verse>
  <!-- weitere Verse... -->
</text>
<!-- Melodie fehlt noch. -->
<melody />
</song>
```

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Video hierarchy I

- Digital videos are composed of the following elements:

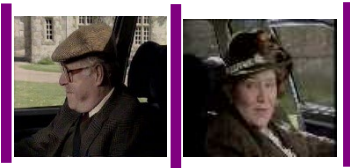
- Frame



- Shots



- Shot boundaries



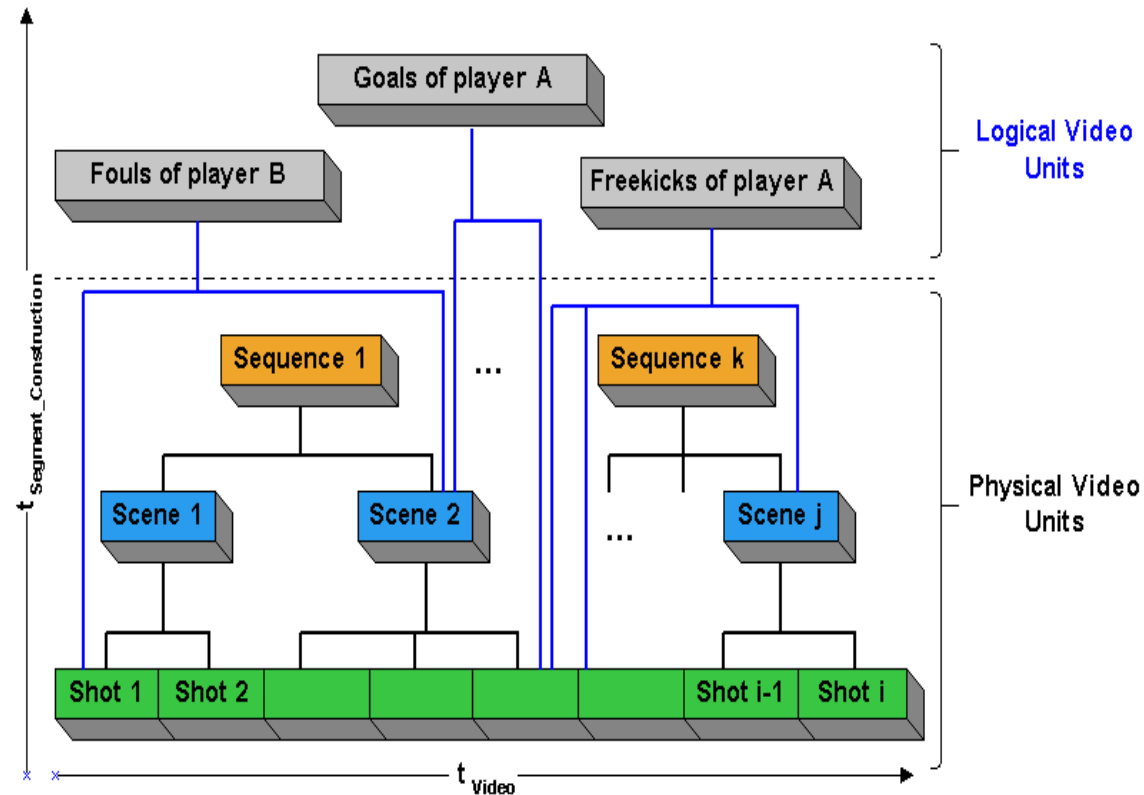
- Scenes



Video hierarchy II

- Partition of a video stream in physical and logical video segments

- The higher the level of a video unit, the higher the amount of necessary semantic information
- Logical units are based on semantic content



Video hierarchy

Shot

- Characteristics
 - General Shot
 - The object is relatively far away from the camera.
 - Medium Shot
 - The object is quite close to the camera.
 - Close Up
 - The object is very close, it almost represents the whole image.
- Types
 - Static
 - The camera does not move for the duration of the shot!
 - Dynamic
 - The camera position changes during the shot: zoom, panning, etc.



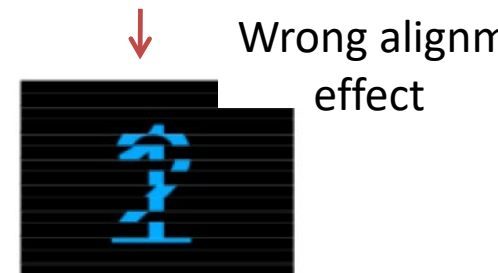
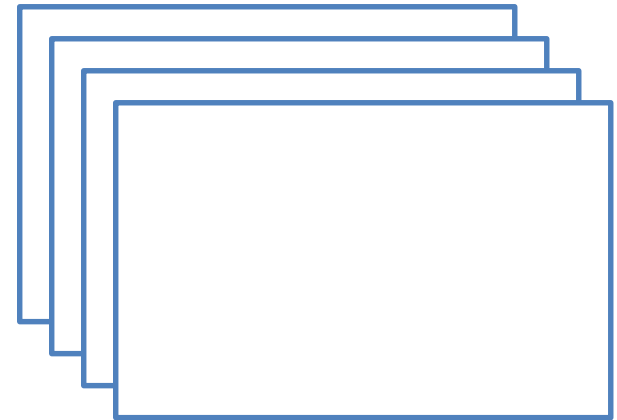
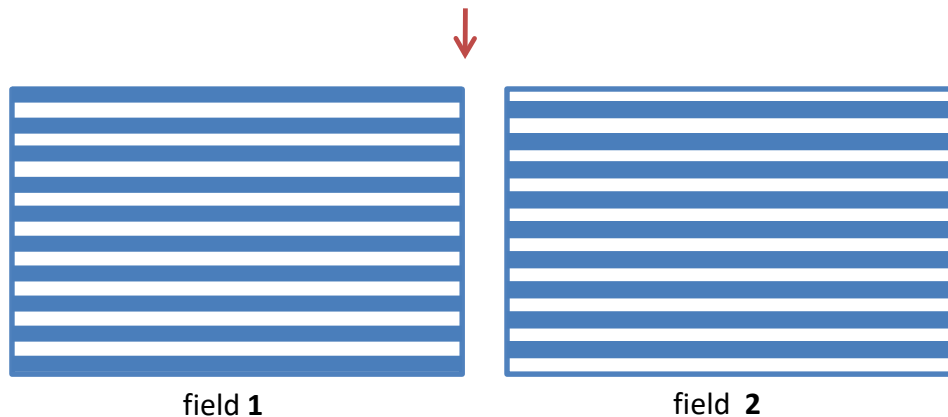
Video hierarchy

Scene

- Scene (Scenario)
 - Sequence of shots that are coherent in the time and space dimensions with respect to the real or represented world.
 - Scene detection is a subjective task
 - Depends on cultural background, professional training, intuition, etc.
- Screenplay
 - Describes all scenes, their dialogs as well as the setup of the cameras.

Video Frame Rate

- A video is a sequence of frames
- It has a specific resolution and must be played with a specific frame rate.
 - e.g. 25 fps, 30 fps (29.97), 50 fps
 - progressive (p) or interlaced (i)
 - Interlaced Frames are also called fields



Usual Resolutions

- **576i** (digital equivalent to PAL)
 - 720x**576** Pixel, **50 half-images** per second
 - Video-DVD, DVB-S, DVB-T (PAL e.g. UHF/VHF)
- **720p** (HDTV)
 - 1280x**720** Pixel, **60 images** per second
 - Blu-ray video, DVB-S2 (*HD Ready*):
 - e.g. ORF HD, ARD/ZDF HD
- **1080p/i** (HDTV)
 - 1920x**1080** Pixel, **24/50/60 images (p)** or **50/60 half-images (i)** per second
 - Blu-ray video, DVB-S2 (*Full HD*):
 - e.g. Servus TV HD, Astra HD
- **2K, 4K** (Digital Cinema)
 - 2048x1080 Pixel and 4096x2160 Pixel, **24 images** per second
 - Digital cine films
- **4320p** (UHDTV)
 - 7680x**4320** Pixel, 60 images per second
- **8K UHD** (7680 × 4320 progressive scan)

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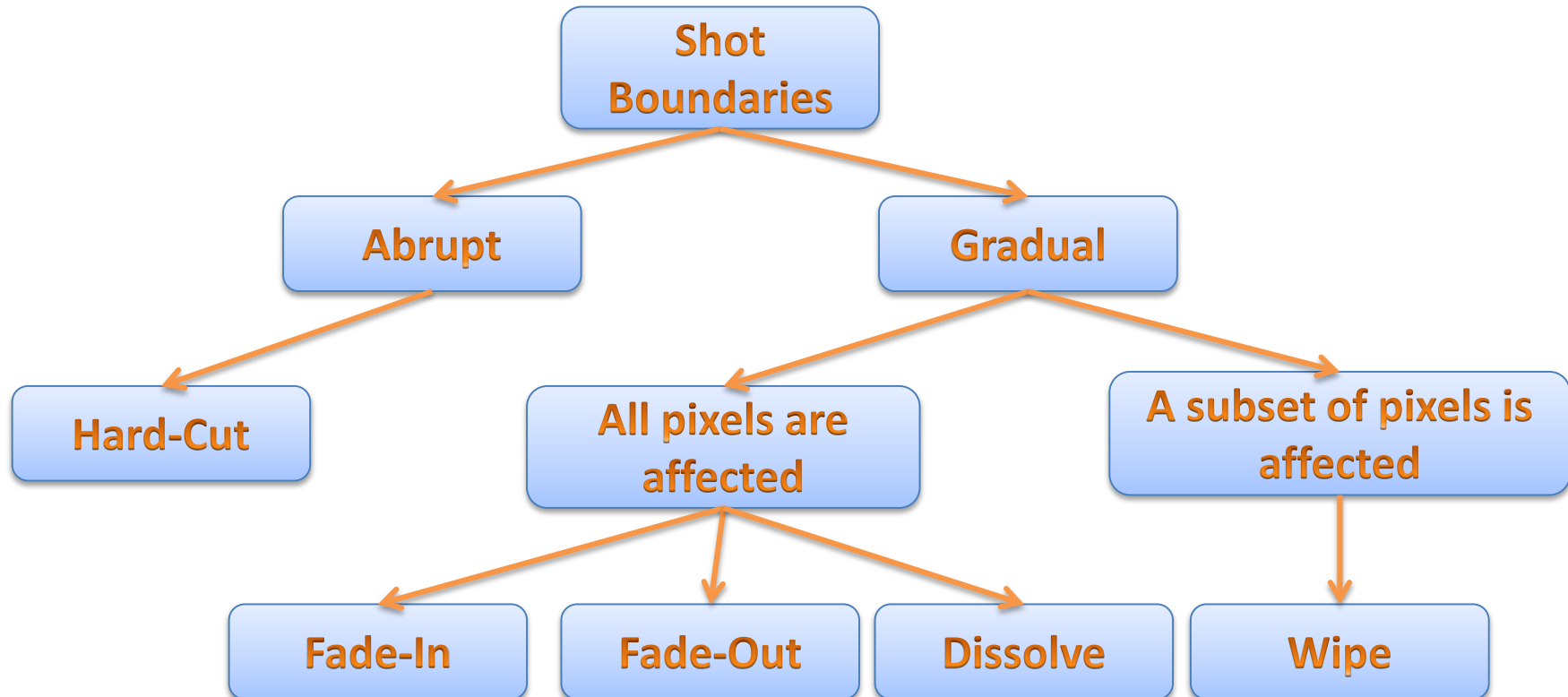
Shot Segmentation

- **Challenges**
 - Object motion
 - Ex.: A person moves within a camera shot, ...
 - Camera motion
 - Ex.: Zoom, panning, ...
 - Lighting changes
 - Camera blitz, illumination, ...
 - Different types of shot boundaries
 - Ex.: Fade, Dissolve, ...
- Preventing *False positives*
 - Adjust threshold values
 - Empiric restrictions
 - Ex.: A shot must last more than 100 frames.

Shot Segmentation

Overview

- Classification according to spatial/temporal properties



Shot Segmentation

Shot types

- **Hard-Cut**
 - Sudden change of the whole image content

Cut



Shot-Segmentation

Shot Types

- **Fade-In**
 - Change from a black image to a complete image
- ▶ **Fade-Out**
 - Change from a complete image to black

Fade-Out

Fade-In



Shot-Segmentation

Shot Types

- **Dissolve**
 - Mix of two shot sequences where the first one is faded out while the second one is faded in.



Shot-Segmentation

Shot Types

- **Wipe**
 - A new shot "pushes away" the previous one through a horizontal/vertical/form-based movement.



Vertical motion

Shot-Segmentation

Spatio-temporal properties

Types of shot boundary		Cut	Wipe	Dissolve	Fade
Spatial	Content change in the whole frame	yes	no	no	yes
	Content change in a subset of pixels	no	yes	no	no
	Brightness change to black	no	no	no	yes
Temporal	Sudden change of content between consecutive frames	yes	no	no	no
	Slow change of content over several frames	no	yes	yes	yes

Evaluation of Shot Detection Algorithms

- **Recall**

- Ratio of the number of detected Shots (S_D) to the total number of shots (detected S_D and undetected S_M)

$$R = \frac{S_D}{S_D + S_M}$$

- ▶ **Precision**

- Proportion of correctly detected shots (S_D corresponds to the number of detected shots and S_F to the wrongly detected shots (*false positives*)).

$$P = \frac{S_D}{S_D + S_F}$$

Shot detection methods

Classification

- **Uncompressed domain**

- Video must first be decompressed
- The methods make use of raw data
 - Pixel-based
 - Histogram-based
 - ...

- **Compressed domain**

- The methods use properties of the compressed data (e.g.: motion vectors, etc.)
 - Macro-block-based
 - Motion vectors
 - ...

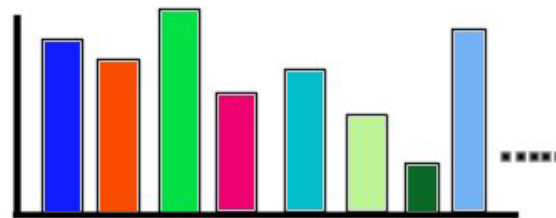
- Many methods are listed at:

- <http://www.visionbib.com/bibliography/applicat821.html>

Shot Detection Methods

Color histogram

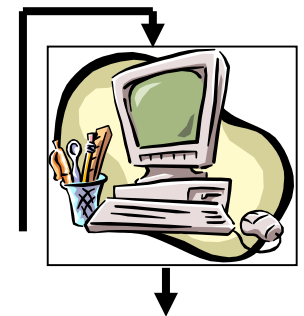
- The **color distribution** of each frame gets extracted
- The distributions are compared with those of each neighbor frame
- If the difference lies above a given threshold
 - -> **Shot**



Color histogram
computed



Comparison with
previous Color histogram



Difference >
Threshold => Shot
change

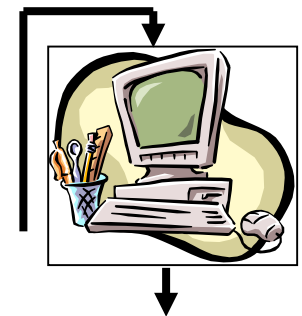
Shot Detection Methods

Edge Detection

- Reduction of the frame colors to greyscale
- Application of an edge detection algorithm
 - Canny Edge Detector, Sobel, etc.
- Differences are computed
 - $\text{Difference} > \text{Threshold} \Rightarrow \text{Shot change}$



Comparison with the
edges of the previous
frame



$\text{Difference} >$
 $\text{Threshold} \Rightarrow \text{Shot}$
 change

Shot Detection Methods

Macroblock (compressed domain)

- Method based on **MPEG compression**
- Frame divided in fixed regions => Macroblocks
- Types of macroblocks
 - I: coded independently from other macroblocks.
 - P: does not encode the region, but rather a motion vector and an error block wrt. the previous frame.
 - B: same as P, but motion vector and error block encoding done wrt. to the previous and next frame

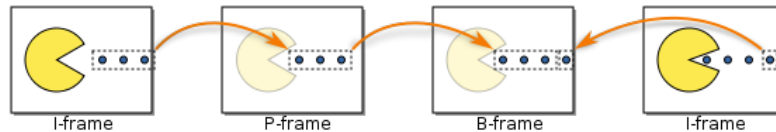
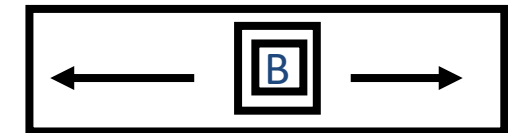
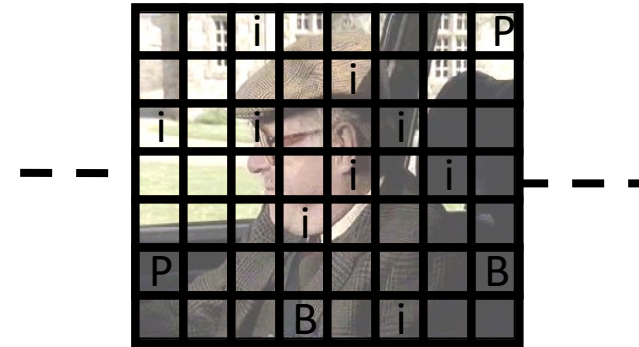


Image source: <https://de.wikipedia.org/>

- Shot boundaries are identified by the **apparition of a specific amount of macroblock types** in a frame.

Frame mit Makroblocks

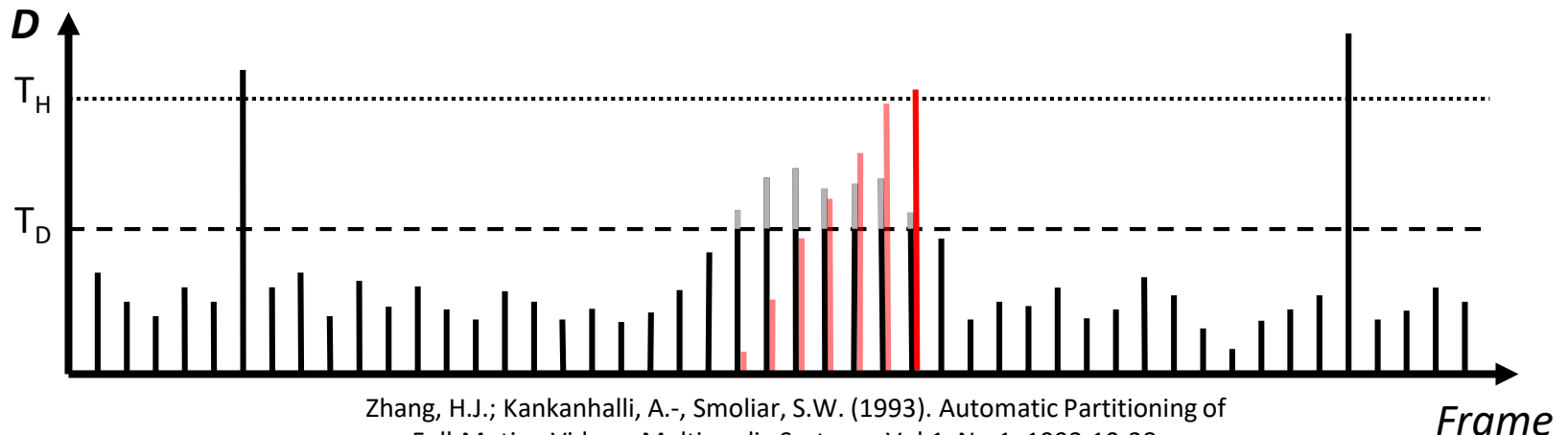


Shot Detection Methods

Gradual means: *twin-comparison*

- Comparison 1: Difference between two consecutive frames
- Comparison 2: Cumulative difference over a sequence of frames
- Difference value:

$$D_{cut} = \sum_{x,y} |I(x, y, t) - I(x, y, t + 1)|$$



Zhang, H.J.; Kankanhalli, A., Smoliar, S.W. (1993). Automatic Partitioning of Full-Motion Videos. Multimedia Systems, Vol.1, No.1, 1993:10-28

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Key Frame

- **Properties:**
 - Should be representative of the content of the shot
 - In the compressed domain, an I-Frame to summarize a video
- **Extraction methods**
 - Optimal algorithm
 - Compares each frame with all the others of a Scene/Shot.
 - The frame with the lowest difference to all others is selected.
 - Very computationally intensive, unusable for most applications.

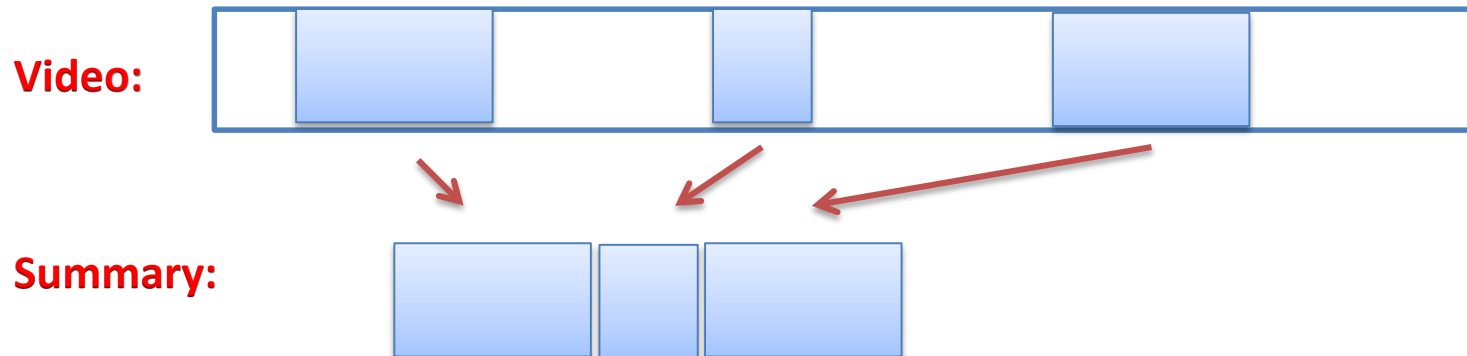
Key Frame

Extraction Methods

- Selection of the **first frame** of a shot/scene as key frame
 - Basically a logical choice since all the other frames of the shot/scene should actually be a continuous extension of/based on that frame.
 - However, the first frame is not necessarily the best and most meaningful representation of a shot/scene.
- Select the frame with the *most complex* or *clearest* content.
 - e.g.: Frames containing text, frames containing easily recognized persons, etc.

Video summarization

- Goal of video summarization:
 - Give a **quick overview** of the content of a video to a user



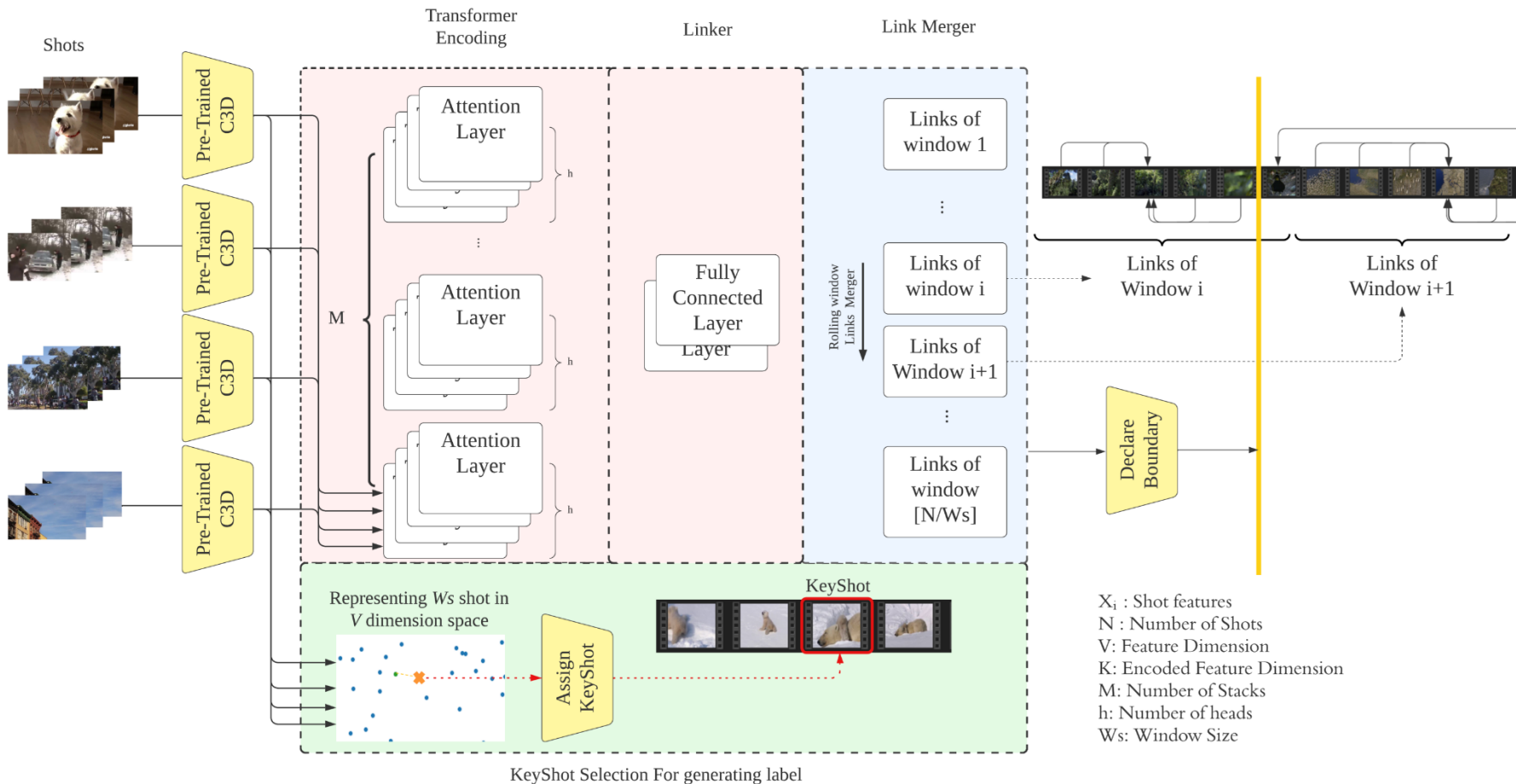
- ▶ Requirements of video summarization:
 - Conciseness => the summary cannot be longer than the Video
 - Content representation=> Temporal and semantic aspects of the content must be preserved
 - Coherence

Video summarization

Classification

- **Types**
 - Static and dynamic methods
- ▶ **Applications**
 - Video Skimming
 - Video Story Board
- ▶ **Two classes** of video summarization
 - Independent
 - Creates a preview representative of the whole content.
 - Depends only on a time limit L
 - Dependent
 - A user has specific preferences wrt. the content (e.g.: specific person, specific event, specific time domain, etc.)

Scene Detection Method



Shu-Ming Tseng, Zhi-Ting Yeh, Chia-Yang Wu, Jia-Bin Chang, Mehdi Norouzi. 2023.
 Video Scene Detection Using Transformer Encoding Linker Network (TELNet). In *Sensors* 2023, 23(16), 7050;
<https://doi.org/10.3390/s23167050>

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Video Formats

Differentiation: Format - Codec

- A **file format** represents a kind of „architecture“
- Special notation about the way a file is structured (programs that read or create header-data, storage of audio-data, etc.)
- Examples of video formats: **MPG, VOB, AVI, ASF**
- Different compression methods (Codecs) possible
- A file with .AVI extension may also contain an MPEG-1 encoded video

Video formats

WebM



- **WebM** is an **open** standard for the distribution of media files over the Internet.
- Provided by Google as **free** software under a BSD-license.
- Based on the Video-Codec **VP8** (developed by On2 Technologies) and the **Vorbis** Audio Codec
- Available as free codec for the HTML5 Video tag

Overview of container formats

Typische Kombinationen			
Container	Name	Videocodecs	Audiocodecs
3GP/3GP2	3rd Generation Mobile	H.263, MPEG-4, H.264	AMR-NB/WB, (HE-)AAC
AVI	Audio Video Interleave	MPEG-4, DV, MJPEG, Indeo, Cinepak	MP3, MP2, (AD)PCM, AC3
ASF	Advanced Streaming/Systems Format	Windows Media Video, VC-1, MS MPEG4 v3	Windows Media Audio
DIVX	DivX Media Format	DivX	MP3, AC3, PCM
DV	Digital Video	DV	PCM
DVR-MS	Microsoft Digital Video Recording	MPEG-2	MP2, AC3
EVO	Enhanced VOB	H.264, VC-1, MPEG-2	(E)AC3, DTS (HD), PCM
FLV	Flashvideo	H.263, VP6, H.264	MP3, AAC, ADPCM
M2TS/MTS	MPEG-2 Transport Stream (192 Byte)	H.264, VC-1, MPEG-2	(E)AC3, DTS (HD), PCM
MKV	Matroska	H.264, MPEG-4	MP3, AC3
MP4	MPEG-4	MPEG-4, H.264	AAC
MPG	MPEG Program Stream	MPEG-1, MPEG-2	MP2
MOV	QuickTime Movie	H.264, MPEG-4, MPEG-1, MJPEG, Sorenson Video	MP3, AC3, PCM
OGM	Ogg Media	Ogg Theora, Xvid	Ogg Vorbis, MP3, AC3
PS	MPEG-2 Program Stream	MPEG-2	MP2, AC3, DTS, PCM
RM(VB)	Real Media (variable Bitrate)	Real Video	Real Audio, AAC
TS/TP/TRP/PVR/VDR	MPEG-2 Transport Stream (188 Byte)	MPEG-2, H.264	MP2, AC3
VOB	Video Object	MPEG-2	AC3, DTS, PCM, MP2
WMV	Windows Media Video	Windows Media Video, VC-1	Windows Media Audio

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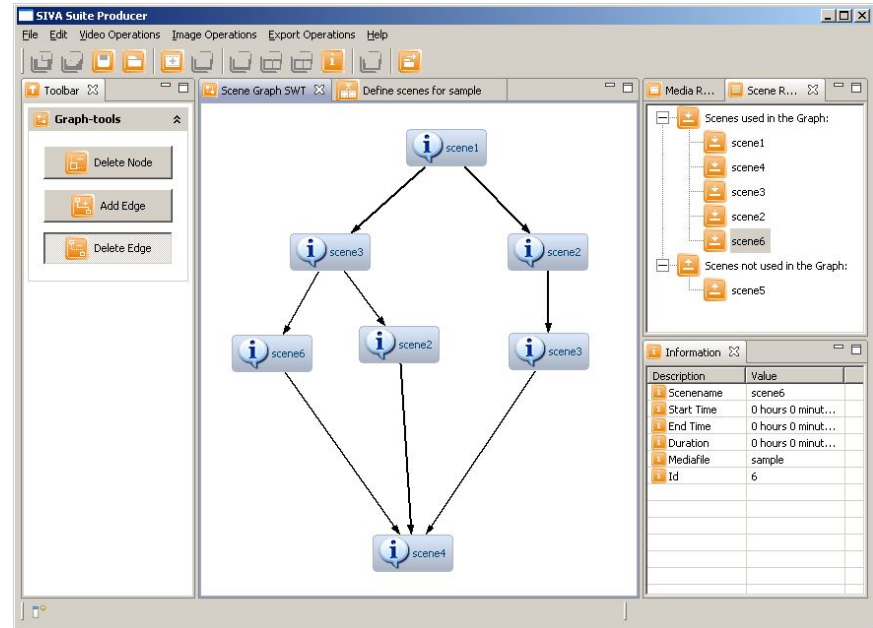
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Interactive videos

- An interactive video is a **digitally enhanced** form of video material, which provides viewers with additional information about the content and lets them control **the flow of video content in particular through advanced navigation options**.
- Arrangement
 - Scene Graphs



Example project

- SIVA Project:

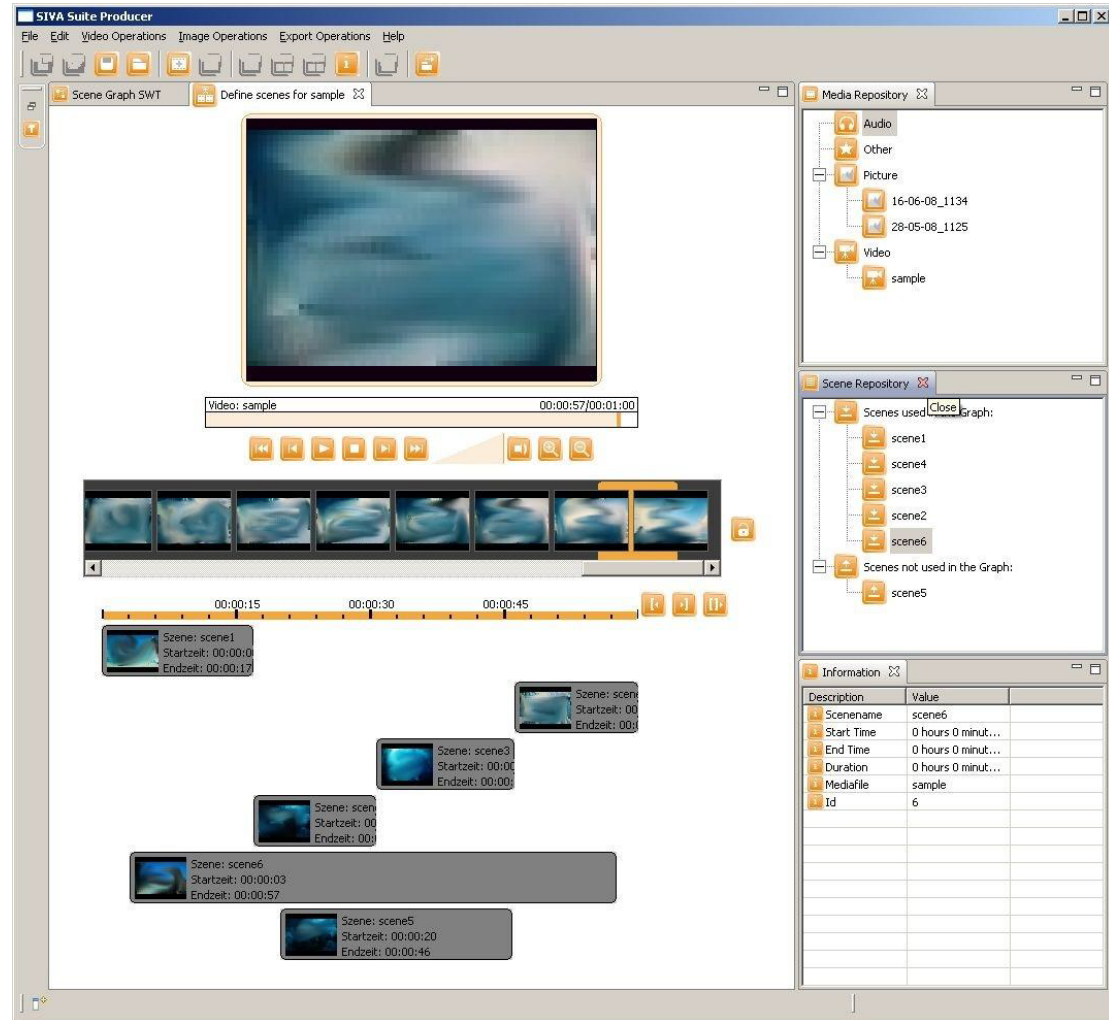


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Audio coding - Basics

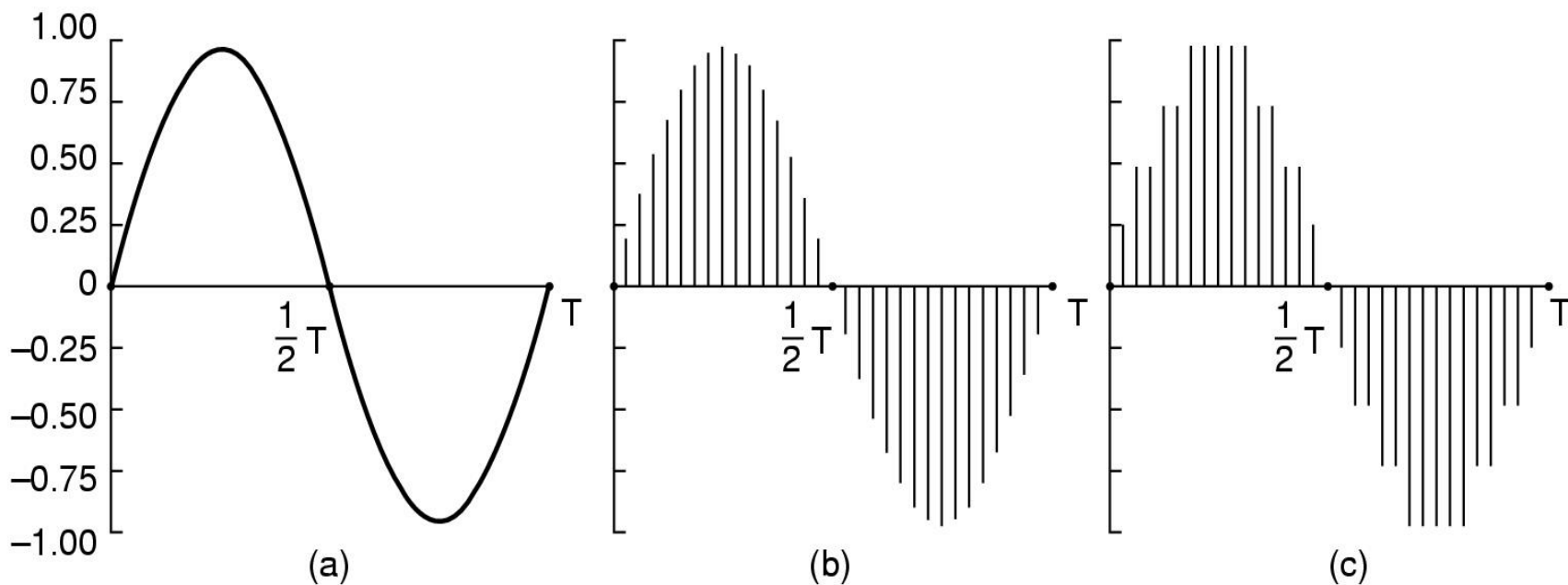
- **Audio (acoustic) waves**
 - One-dimensional, acoustic (pressure) wave
 - Causes vibration of the eardrum (human perception) or of a microphone
- **Frequency spectrum of the human ear**
 - 20 – 20.000 Hz (20 KHz)
 - Quasi logarithmic perception: the ratio of amplitudes A and B is expressed by: $\text{dB} = 20 \log_{10} (A/B)$

Very low pressure (20 μ Pascal)	0 dB
Conversation	50 – 60 dB
Heavy traffic	80 dB
Rock band	120 dB
Pain threshold	130 dB

Analog-Digital Transformation (ADC)

- Sampling of the audio waves each ΔT seconds
 - If the acoustic wave is a linear superposition of undisturbed sine waves with max. frequency f :
 - Sampling rate= *Shannon and Nyquist Theorem*
 - Uniform sampling of rate of at least $2*f$, in order to be able to reconstruct the original signal without information loss.
 - e.g. CD are sampled at 44.1 KHz $\approx 2 * 20$ KHz
- Quantization
 - Precision of the digital sampling process depends on the number of bits
 - Quantization error
 - Error due to the finite number of available bits/sample

Audio coding example



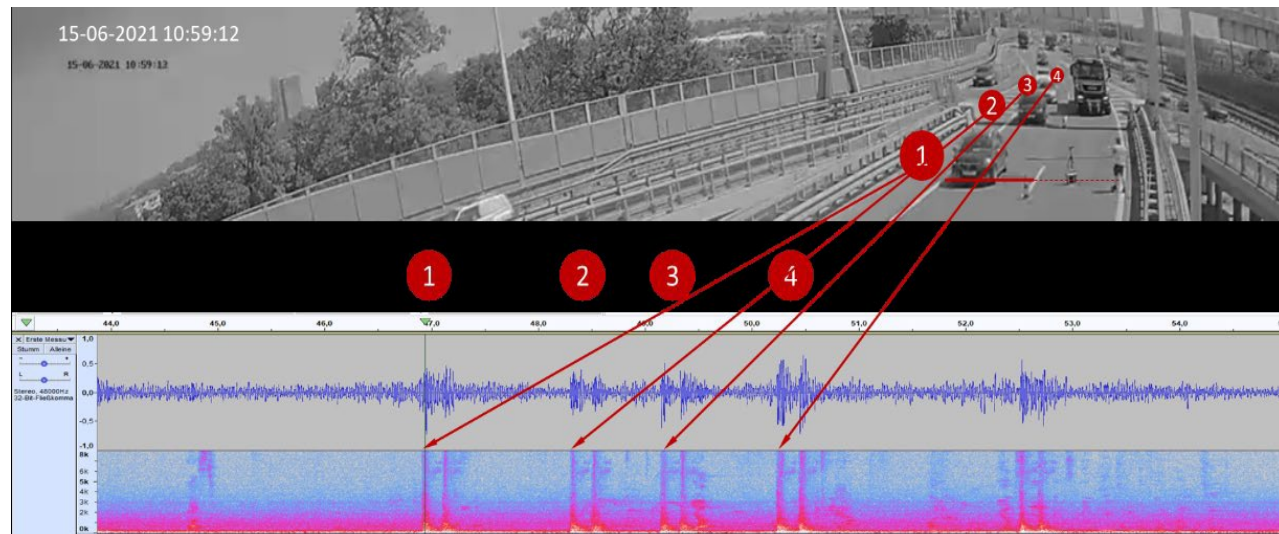
- a) Sinus wave
- b) Sampling of the sinus wave
- c) Quantization of the sampled data to 4 Bits

Vehicle Classification based on Audio Signal

- Research project in cooperation with Bernard Technology
- Audio Sensor installed under a Bridge and test data has been recorded in combination with synchronized video data
- Evaluated Random Forest [1] and XGBoost [2] Method which resulted in 92% accuracy (XGBoost)

[1] Leo Breiman, Random forests. In Machine Learning, 45(1), 5-32, Springer.

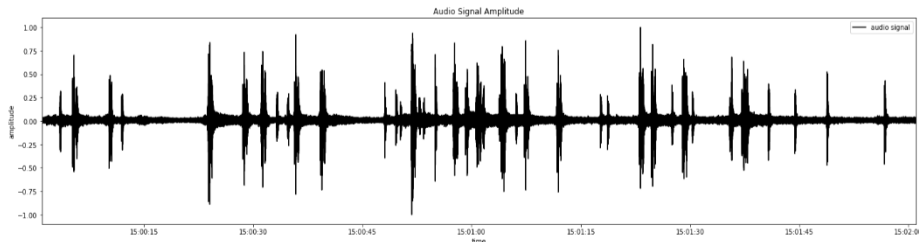
[2] Tianqi Chen, Carlos Guestrin, XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785-794, 2016, DOI: 10.1145/2939672.2939785



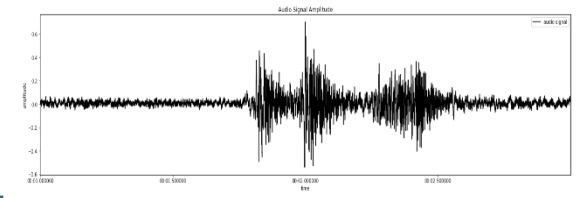
Vehicle Classification based on Audio Signal

- Creation of Ground Truth according to synchronized video data and Label Studio (<https://labelstud.io/>)

Audio-File



Peak Detection & Snippet Creation



Label Studio Implementation

2. Edit Labeling config

```
6 </Labels>
7
8 <AudioPlus name="audio" value="$audio"/>
9 <View style="padding: 10px 20px; margin-top: 2em; box-shadow: 2px 2px 8px
10   visibleWhen="region-selected">
11   <Header value="Provide transcription" />
12   <TextArea name="transcription" toName="audio"
13     rows="2" editable="true" perRegion="true"
14     required="true" />
15 </View>
16 <View style="padding: 10px 20px; margin-top: 2em; box-shadow: 2px 2px 8px
17   visibleWhen="region-selected">
18   <Header value="Select Gender" />
19   <Choices name="gender" toName="audio"
20     perRegions="true" required="true">
21     <Choice value="Male" />
22     <Choice value="Female" />
23 </Choices>
24 </View>
25
26 <View style="width: 100%; display: block">
27   <Header value="Select region after creation to go next"/>
28 </View>
29
30 </View>
31 </View>
```

3. Inspect Interface preview



Labeled Data

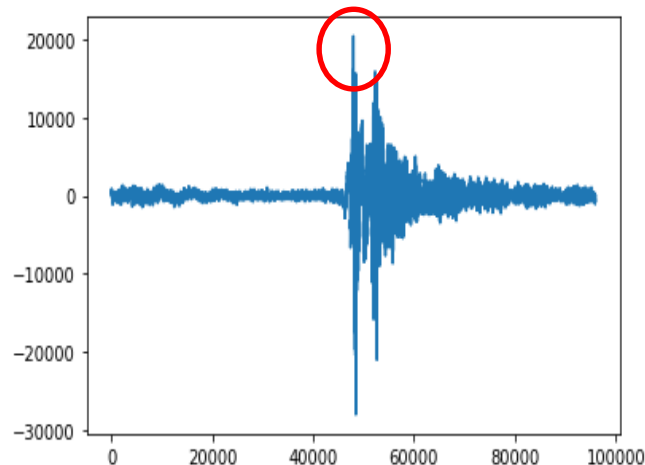
```
[
  {
    "original_length": 3.774375,
    "value": {
      "start": 0,
      "end": 0.6394089226973688,
      "labels": [
        "Speaker 1"
      ]
    },
    "id": "wavesurfer_8aaohk6v8a",
    "from_name": "label",
    "to_name": "audio",
    "type": "labels"
  }
]
```

Start typing in the config, and you can quickly preview the labeling interface. At the bottom of the page, you have live serialization updates of what Label Studio expects as an input and what it gives you as a result of your labeling work.

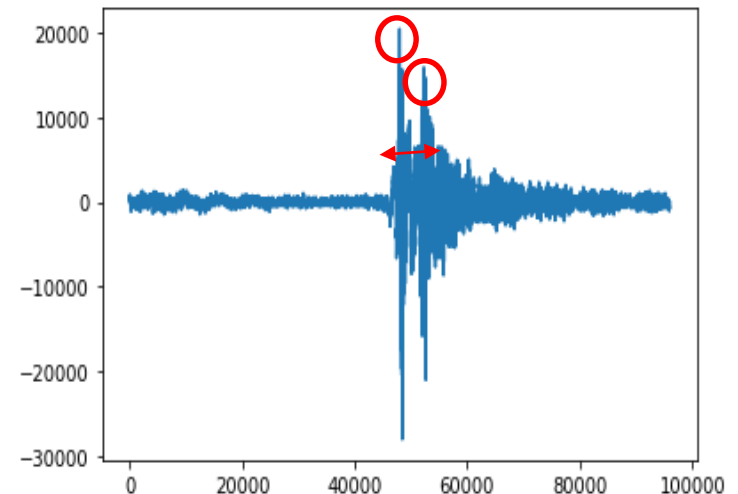
Vehicle Classification based on Audio Signal

- Extraction of statistical features by using SciPy (<https://scipy.org/>) for training.

Max Peak:



Temporal distance between peaks



The End